1. Introduction

The goal of this project is to analyze and forecast building energy consumption using machine learning methods to support sustainable development goals, particularly UN Sustainable Development Goal 7 (Affordable and Clean Energy).

To enhance the originality of the study and better understand how model performance varies depending on the data structure, two different datasets were used.

Accurate energy consumption forecasting can facilitate efficient resource management and reduce environmental impact.

2. Methodology

The study used two datasets containing environmental and operational variables such as temperature, humidity, occupancy level, heating, ventilation, and air conditioning (HVAC) usage, lighting usage, renewable energy, and calendar characteristics.

Prior to modeling, the datasets were preprocessed by converting numerical columns to appropriate formats, removing corrupted rows, and handling missing values. Categorical variables were converted to numerical form using one-hot encoding.

Furthermore, for the second dataset, temporal characteristics such as hour and day were extracted from timestamps to identify temporal patterns.

Both datasets were split into training (80%) and test (20%) sets.

Two regression models were applied:

- Linear Regression (baseline model)
- Random Forest Regressor (ensemble-based model)

Feature scaling (StandardScaler) was applied to the Linear Regression model.

3. Results

Model performance was assessed using the mean absolute error (MAE), root mean square error (RMSE), and R^2.

In the first dataset, linear regression performed slightly better than random forest, achieving a lower RMSE and a higher R^2:

Linear regression: RMSE = 5.06, R^2 = 0.61 Random forest: RMSE = 5.42, R^2 = 0.55

However, in the second dataset, random forest performed significantly better than linear regression:

Random forest: RMSE = 67.81, R^2 = 0.54 Linear regression: RMSE = 91.13, R^2 = 0.17

These results indicate that model performance varies across datasets and depends on the underlying data structure.

4. Interpretation of Results

The results show that model performance varies depending on the dataset and underlying data structure.

For the first dataset, linear regression performed better, indicating that the relationship between features and energy consumption is largely linear and contains limited nonlinear patterns. In contrast, for the second dataset, random forest performed significantly better than linear regression. This suggests the presence of more complex and nonlinear relationships that are better captured by ensemble models. Feature significance analysis also confirms these findings. In the first dataset, temperature was identified as the most influential factor influencing energy consumption. However, in the second dataset, temporal features (e.g., clock) were found to be the most significant variables. This indicates that energy consumption is influenced not only by environmental conditions but also by temporal patterns of use and user behavior.

5. Limitations and Improvements

This study identified several limitations. The first dataset is relatively small (~1000 observations), which may limit the generalization ability of the model. Furthermore, the first dataset did not exploit temporal patterns, while the second partially incorporated them through feature engineering.

Furthermore, model performance could be further improved, particularly for the random forest model, which was used without hyperparameter tuning. Future improvements could include hyperparameter optimization (e.g., GridSearchCV), the use of advanced models such as XGBoost, and the inclusion of more comprehensive time series analysis. Collecting larger and more diverse datasets would also improve the model's performance and generalization ability.

6. Comparison of Two Datasets

Two different datasets were used to predict energy consumption with Linear Regression and Random Forest models. The first dataset includes building-related variables such as temperature, humidity, square footage, occupancy, HVAC usage, lighting usage, holiday status, and renewable energy. The second dataset focuses on appliance energy consumption and includes environmental variables together with time-based features such as hour and day.

The comparison shows that the structure and complexity of the datasets directly affect model performance. In the first dataset, the relationship between the features and energy consumption is more regular, so Linear Regression can give reasonable results. However, Random Forest can also improve prediction by capturing interactions between different variables.

In the second dataset, energy consumption has more complex and nonlinear patterns because appliance usage changes depending on environmental conditions and time. For this reason, Random Forest is more effective than Linear Regression on this dataset. Overall, this comparison shows that choosing the correct model according to the characteristics of the dataset is important for accurate energy consumption prediction and sustainable energy management.

7. Conclusion

This study demonstrates that machine learning models can effectively predict energy consumption and support sustainable energy management.
A comparison of two different datasets reveals that model performance depends on the structure and complexity of the data. Linear regression performed better on a dataset with linear relationships, while random forest showed superior results on a dataset with more complex and nonlinear patterns.
These results highlight the importance of selecting appropriate models based on data characteristics and support the role of artificial intelligence in achieving sustainable development goals.